Simulating Sgraffito on Layered Materials

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***Abstract*—Robotic sgraffito is a contact-rich fabrication task where material is selectively removed to reveal patterns across multiple layers. Existing approaches to robotic sgraffito use separate, tool-specific pipelines, making it difficult to reuse manipulation skills across different tools and tasks. This project uses a simulation and planning framework, developed by our supervisor, to study these skills in a shared environment. The system supports parameterized canvas and tool configurations, three action primitives (scrape, trace, and dot), an ergodic planner for generating scrape and dot actions from target images, and both geometric CSG and physics-based simulation. Our experiments show that increasing contact force from 2 N to 12 N and commanded scrape depth from 0.8 mm to 2.5 mm both produced results approaching an approximately 0.9 mm depth ceiling, suggesting a limit of the simulated material response. We also show that penetration depth can control which colour layer is revealed and that the ergodic planner can generate a manipulation mission from an input image. This framework provides a physically grounded foundation for translating artistic intent into reusable robotic sgraffito actions.**

I. Introduction

Sgraffito is a decorative technique where layers of differently coloured plaster are selectively removed to reveal patterns and designs underneath. The technique was popularized during the Italian Renaissance in the 15th and 16th centuries, where it was used to decorate walls and other architectural surfaces. Creating detailed sgraffito requires specialized skill and can be labour-intensive, which makes it a useful application for robotic fabrication. It also provides a useful testbed for contact-rich manipulation, since a robot must combine different actions while controlling its interaction with the material, including tool contact, material removal, and tool-material behaviour.

Robotic manipulation tasks can be broken down into specialized skills, and recent work has explored learning these skills from demonstrations and videos [1], [2]. However, these approaches generally learn behaviours for individual tasks rather than reusable primitives that can be combined across different tasks. Language-based approaches have also been explored for breaking down and composing robot tasks [3], [4]. However, language models alone cannot capture the physical interactions involved in contact-rich fabrication, such as contact forces, material removal, and how different tools interact with the material. Robotic sgraffito has also been explored directly by Lyu et al. [5], but their methods were developed as separate pipelines for each tool, without a shared representation or quantitative calibration of the resulting material changes. This motivates the need for a physically grounded simulation environment where manipulation primitives can be defined and evaluated before they are learned or composed into larger tasks.

In this work, we use a simulation and planning framework, developed by our supervisor, to study how different tools and manipulation actions affect layered materials. The framework includes (1) a geometric simulation approach for layered material removal, together with an initial physics-based solver undergoing calibration, (2) configurable canvas and tool representations that support different geometries, tilt angles, and interaction primitives such as scraping, tracing, and dotting, (3) an ergodic planning component that converts a target image into a sequence of scrape and dot placements by matching a target coverage density, and (4) an export pipeline for interactive inspection of simulated outcomes. We use this framework to design and run a set of experiments characterizing how these primitives behave under different tool and material conditions, providing a foundation for future work on learning reusable manipulation primitives and extending automated planning to the full set of interaction types, as shown in Fig. 1.

II. Related Literature

A. Data-driven skill acquisition for manipulation

Recent work has explored how robots can learn manipulation skills directly from demonstrations rather than relying on manually designed behaviours. Our long-term goal is to build a system for robotic sgraffito that can learn reusable primitive motions from demonstrations.

Hou et al. [1] proposed a system that learns manipulation policies from kinesthetic demonstrations, jointly predicting the robot's motion and compliance. Their approach was demonstrated on tasks such as object flipping and vase wiping. However, the learned representation is a task-specific policy rather than a reusable primitive that can be independently combined with other skills.

Another direction is learning manipulation behaviours directly from videos. Liu et al. [2] proposed a method for imitating human behaviours from raw video while adapting to different contexts. This approach could reduce the need for direct robot demonstrations and provide a way to learn manipulation behaviours from more readily available human

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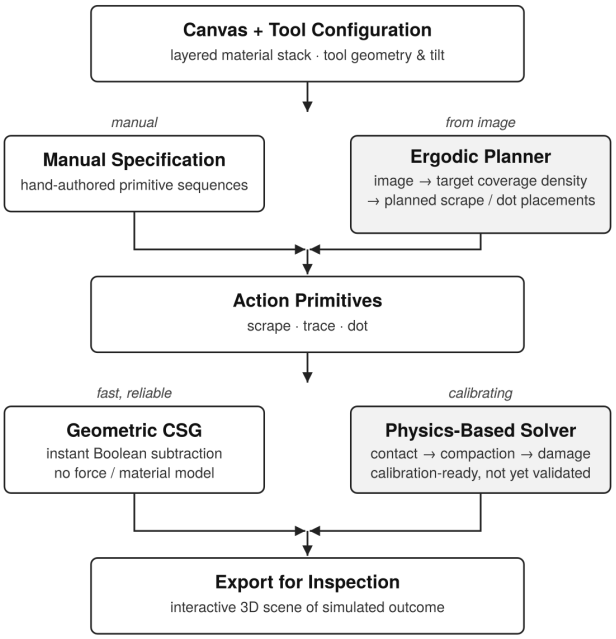


Fig. 1. System overview. Primitives can be specified manually or planned automatically from an image via the ergodic planner (scrape and dot only); simulation proceeds through either the fast geometric solver or the physics-based solver, currently undergoing calibration.

demonstrations. However, similar to the approach of Hou et al., the resulting behaviour is still tied to the demonstrated task rather than being represented as a set of reusable primitives.

Rather than extracting primitives from demonstrations, our project uses existing simulation infrastructure to define and evaluate potential manipulation primitives for robotic sgraffito. This provides a foundation for future work on learning reusable primitives from demonstrations.

B. Robotic fabrication of sgraffito

Robotic sgraffito has been previously explored by Lyu et al. [5], who developed RoboSgraffito, a robotic fabrication project that investigated multiple approaches to producing sgraffito artwork. RoboSgraffito is the closest precedent to our project, as both use robotic interaction with layered material for art.

RoboSgraffito implemented five fabrication methods: grinding, planing, pulling, stamping, and scratching. Each method used a different end-effector and image-to-motion mapping. Grinding used repeated abrasion, with stroke density, overlap, and tool angle controlled by image darkness. Planing used a chisel with repeated push motions and crosshatched paths to create engraved and geometric patterns. Pulling used a flexible blade whose path density controlled the amount of material removed, while stamping used point-based tool placements to create tonal images by displacing material. Scratching used a triangular tool whose groove depth and angle were mapped to pixel brightness. Together, these methods demonstrated several ways of translating digital images into physical surface patterns.

However, each method was developed as a separate pipeline around its specific tool and material interaction. There was no shared representation across the different tools, making it difficult to transfer an approach between methods or reuse the same manipulation primitive with a different end-effector.

The project also did not establish a quantitative relationship between tool parameters and physical outcomes. Factors such as tool angle, speed, spacing, and material condition could affect the resulting artwork, but these effects were not systematically calibrated. As a result, it is difficult to predict the outcome of a new manipulation strategy before physically testing it.

Our project instead uses a shared simulation infrastructure that can represent different tool geometries and interaction types, including scraping, tracing, and dotting. This provides a foundation for defining and evaluating reusable manipulation primitives and studying how different tool actions affect the material. Future work can build on this framework to learn these primitives from demonstrations.

III. Problem Formulation

The simulator used in this project takes three main inputs: a layered canvas, a tool, and a sequence of actions. The canvas contains multiple material layers, each with its own colour, appearance, and physical properties. The tool is defined by its geometry and tilt angle. The actions are represented as primitives, such as scraping with a specified direction and depth, tracing along a path, or dotting at a given position and force.

The simulator applies these actions to the canvas and produces the resulting material state. This includes the height of each layer, as well as information about strain, damage, and mass that capture how the material has been physically altered. The final surface can also be exported for interactive inspection.

The simulation makes a few simplifying assumptions. First, the tool is assumed to move slowly, so dynamic or impact effects are not modelled. This matches how sgraffito is normally performed. Second, the material is represented as a 2.5D height field instead of a full 3D model, since sgraffito mainly involves removing material from the surface and does not require overhangs or undercuts. Finally, automated generation of an action sequence from an image is currently limited to scrape and dot primitives, using a separate ergodic planning component (Section IV-C); trace sequences are still specified manually, and extending automated planning to cover all primitive types is left for future work.

IV. Methodology

The framework used in this project is implemented across two repositories: the canvas and tool simulation framework [6], and a companion ergodic planning codebase [7], both developed by our supervisor.

A. Canvas and Tool Representation

The simulation starts with a layered canvas and a tool. The canvas is defined as a stack of material layers, with each layer having a minimum and maximum height, a material type, and a flag indicating whether it can be removed. Each layer also has visual properties such as colour and roughness, as well as physical properties such as friction, abrasion, plastic compaction, and fracture. The canvas is represented internally using one OpenVDB level-set grid per layer, with a configurable voxel resolution. Higher resolution allows for more surface detail but increases computation time.

The tool is represented as a watertight mesh and can either be manually designed or generated from a parameterized tool specification. The simulator currently supports several tool profiles, including ribbon-shaped, square, teardrop, ball, and curved-blade geometries. Each profile has its own geometric parameters, such as width, depth, curvature, or radius.

The geometry and tilt angle of the tool determine the area of material that it can remove. For each tool pose, the simulator calculates this cutting footprint by rasterizing the lower cutting edge of the tool into an XY height field. These footprints are cached for different roll angles so they can be reused during simulation.

B. Action Primitives

The system represents tool interactions using three basic action primitives: scrape, trace, and dot. These primitives are intended to act as basic building blocks that could eventually be combined by a higher-level planner or learned from demonstrations.

A scrape is a straight-line tool movement with a specified heading and depth. A trace follows a curved path made from cubic Bézier segments. The path is resampled by arc length so that the tool follows it at regular intervals. Traces can use either direct CSG subtraction or a footprint-based method that removes material based on the area swept by the tool; the footprint-based method is closer to the behaviour of a real scraping tool because it clears material across the full swept area. A dot is different from the other two primitives, as it does not remove material. Instead, the tool applies a point contact force that compacts and displaces the material.

Currently, the sequence and parameters of these primitives are specified manually by default – for example, as a hand-authored stroke program or a set of Bézier paths. Scrape and dot placements can alternatively be generated automatically from a target image using the ergodic planner described in Section IV-C. Trace sequences, however, are always specified manually; extending automated planning to cover traces as well is left as future work.

C. Ergodic Primitive Placement

In addition to manually specified primitive sequences, the system supports automated placement of scrape and dot primitives using an ergodic coverage planner. A source image is first posterized into per-layer target coverage densities, one for each material layer to be revealed. Each target density is converted into a Fourier-basis representation over the canvas domain using an ergodic metric formulation [8], implemented in a companion codebase developed by our supervisor [7]. A greedy planner then incrementally selects primitive placements – positions for dot actions, and positions and paths for scrape actions – that minimize the ergodic metric against the target density at each step, producing an ordered sequence of placements that can be executed by the simulator described in Sections IV-A–IV-B.

Trace primitives are not currently covered by the planner and remain manually specified, as described in Section IV-B. Scrape and dot missions were planned somewhat differently in practice: dot-based missions consumed an externally generated mission file, while scrape-based missions were generated directly by the planning script used for this project.

D. Physics Simulation

The simulator currently has two approaches for modelling material removal.

The first is geometric CSG, which is the primary and most reliable simulation path used for generating results. The tool and canvas are represented using signed distance fields, allowing the simulator to perform Boolean subtraction between the tool and removable material layers. This approach is fast and directly produces the resulting geometry, but it does not model forces or detailed material behaviour.

The second approach is a physics-based continuum solver, which is currently under development. The solver models the interaction between the tool and material through contact pressure and traction, followed by plastic compaction, shear, material movement, damage accumulation, and eventual material detachment. The solver includes a calibration procedure for fitting parameters such as elasticity, compaction, friction, shear, fracture, wear, and layer interfaces.

The physics solver currently uses estimated material parameters, so its results have not yet been validated against real material behaviour. Its 2.5D representation also limits it to surface-level effects and cannot model phenomena such as curled chips or undercuts.

E. Export for Inspection

After simulation, the resulting canvas geometry is converted into a mesh and exported as a 3D scene file that can be viewed interactively outside the simulation environment. The framework includes exported examples for three different canvas and tool configurations, including variations in canvas layer composition, tool geometry, and voxel resolution.

V. Simulation and Experimental Results

A. Experimental Setup

We evaluated the geometric CSG simulation and the physics-based continuum solver using a shared canvas and tool setup. The tool-tilt and multi-colour reveal experiments (Sections V-B, V-D) used the geometric CSG path, where commanded penetration directly determines the resulting cut depth, while the dot-compaction and scrape-depth experiments

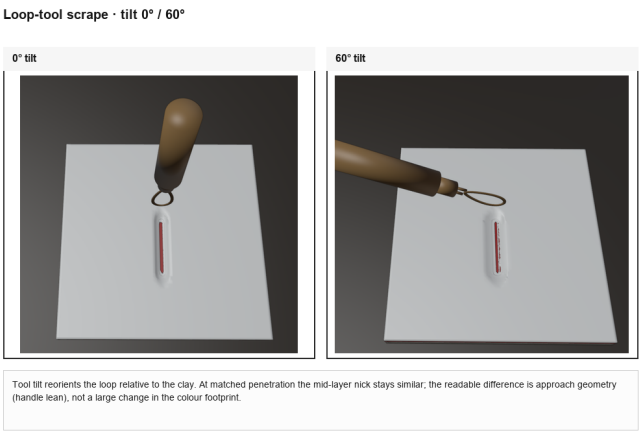


Fig. 2. Loop-tool straight scrape at 0° and 60° tilt. Tilt reorients the tool's approach but produces a similar mid-layer reveal at matched penetration depth.

(Sections V-C, V-E) used the physics-based solver. We tested how tool tilt, contact force, and commanded depth affected the resulting material, and also tested whether different scraping depths could be used to reveal different colour layers. We additionally ran the ergodic planner on a posterized image as a qualitative demonstration of the image-to-mission pipeline.

For the geometric primitives, we tested the loop tool at 0° and 60° tilt using both straight scraping and an S-shaped trace. For the physics-based simulation, we tested the dot tool at contact forces of 2 N, 6 N, and 12 N while keeping the commanded centre depth fixed. We also tested scrape depths of 0.8 mm, 1.5 mm, and 2.5 mm over 16 mm strokes. Each condition was run once, so the results do not include repeated trials or statistical measures such as variance.

B. Effect of Tool Tilt on Geometric Primitives

We first tested whether changing the tool tilt affected the resulting material removal. The loop tool was tested at 0° and 60° tilt using both a straight scrape and an S-shaped trace. At the same penetration depth, both tilt angles produced similar layer reveals. This suggests that, for these tests, changing the tilt mainly changes the way the tool approaches the material rather than substantially changing the resulting colour footprint.

C. Effect of Contact Force on Dot Compaction

We next tested how contact force affects material compaction using the dot tool. The tool was applied at 2 N, 6 N, and 12 N while keeping the commanded centre depth the same. This is a Hertz-contact compaction process only, with no material removed; higher force widens the resulting permanent set but does not make its centre significantly deeper. All three force levels approached a depth of approximately 0.9 mm. Higher forces reached this depth more quickly and produced a wider mark, suggesting that the limiting factor is material compaction rather than the amount of applied force. As the material becomes more compacted, additional force produces less additional depth.

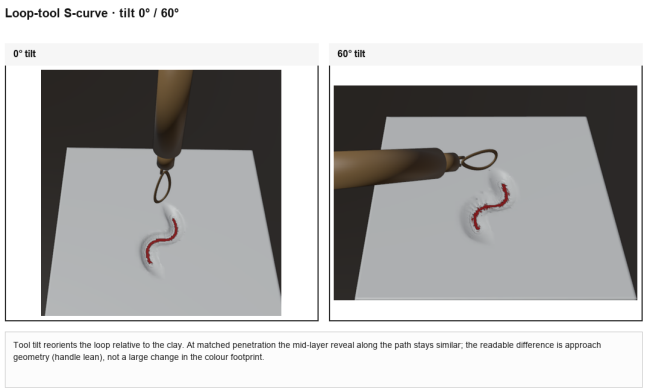


Fig. 3. Loop-tool S-curve trace at 0° and 60° tilt, showing the same tilt-invariance along a curved path.

The three force curves flatten toward the same approximately 0.9 mm depth ceiling, as shown in Figures 5 and 6.

D. Depth-Controlled Multi-Colour Reveal

We also tested whether penetration depth could be used to control which layer of the canvas was exposed. This experiment used the geometric CSG path, where commanded penetration directly determines the depth of the cut, allowing a single pass to reach any layer. Shallow, medium, and deep scraping passes revealed the top white layer, red layer, and lower layer respectively. This shows that different colours can be revealed by changing the commanded depth without changing the tool or contact force. This provides a simple way of controlling the visible layer during a sgraffito operation.

E. Achieved versus Commanded Scrape Depth

We compared commanded scrape depths of 0.8 mm, 1.5 mm, and 2.5 mm over 16 mm strokes using the physics-based solver. Achieved depth tracked the command at first, then saturated near $\sim$0.9 mm. Deeper commands therefore did not produce proportionally deeper cuts. This achieved-depth ceiling is a material/bite limit specific to the physics-based scrape experiment. Figure 7 varies commanded geometric penetration to select which coat is opened, while Figure 8 shows that under the physics-based scrape setup, achieved depth stays near $\sim$0.9 mm even when the command is larger; the two figures describe different simulation paths rather than contradicting each other.

This ceiling is also consistent with the result from the dot-compaction experiment in Section V-C, where different contact forces also approached the same approximately 0.9 mm depth using the same physics-based solver. The fact that two different experiments on the same simulation path produce the same depth limit suggests that this behaviour reflects the physics solver's material response rather than being specific to one particular control parameter.

F. Ergodic-Planned Mission

Finally, we tested the ergodic planner on a flower image as a qualitative demonstration of the full pipeline. The source

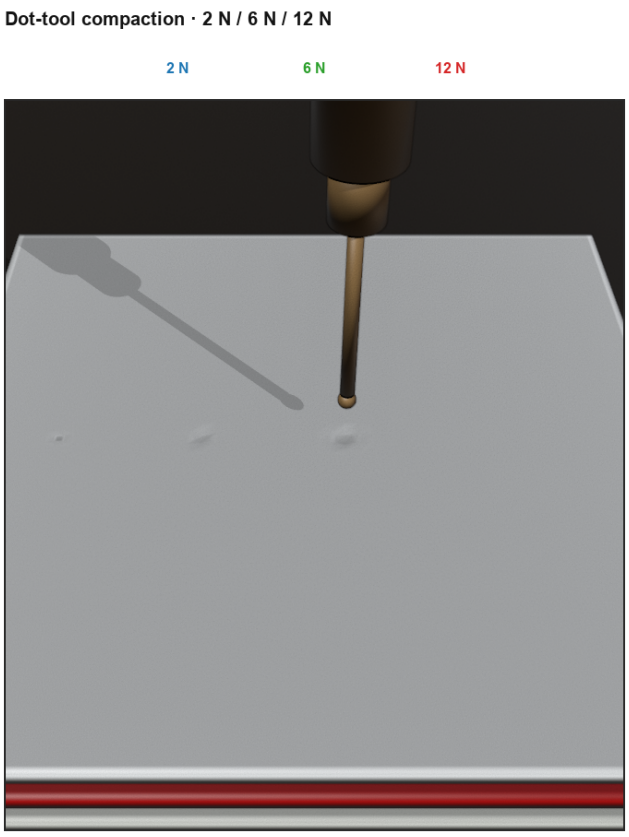


Fig. 4. Dot-tool compaction at 2 N, 6 N, and 12 N and matched commanded centre depth. This is compaction only, with no material removed; higher force widens the permanent set without deepening its centre.

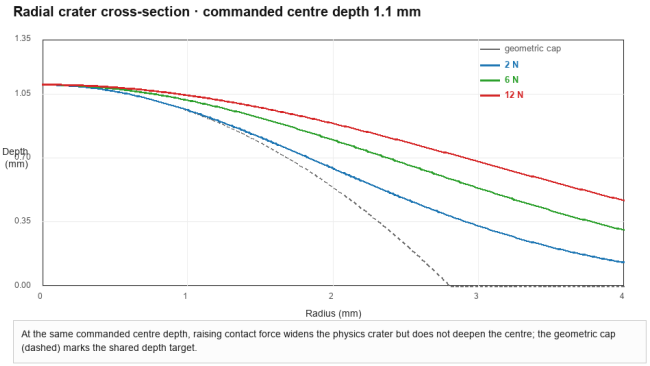


Fig. 5. Radial cross-section through a compacted dot: the geometric cap (dashed) marks the shared commanded depth target, while the three physics-based force curves flatten toward a common ∼0.9 mm ceiling.

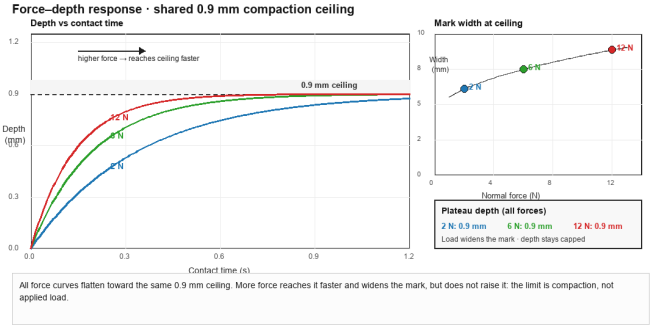


Fig. 6. Depth versus contact time (left) and mark width at the depth ceiling (right), across the three tested forces. Higher force reaches the ∼0.9 mm ceiling faster and widens the mark, but does not raise it.

image was first posterized into a smaller number of colours, after which target density maps were generated for the different material layers. These maps were then used to generate the placement of manipulation primitives. Figure 9 shows the source image, posterized image, and the resulting target masks.

This experiment demonstrates that the system can connect an input image to a planned sequence of manipulation actions. However, this result is presented as a qualitative demonstration rather than a quantitative measure of how accurately the final material matches the input image.

G. Limitations

There are several limitations to these results. First, each experimental condition was run only once, so we cannot report variation across repeated trials. Second, the results cannot be directly compared quantitatively with RoboSgraffito because that work does not provide comparable calibrated measurements. Finally, the approximately 0.9 mm depth ceiling has only been tested with the loop tool, so it is not yet clear whether the same limit applies to other tool geometries.

VI. Conclusion and Future Work

This project used a simulation framework, developed by our supervisor, to provide a physically grounded environment for studying reusable manipulation primitives. The system supports different canvas and tool configurations, three action primitives (scrape, trace, and dot), an ergodic planner for generating scrape and dot placements from target images, and both geometric CSG and physics-based simulation approaches. Our experiments showed that tool tilt had a limited effect on the exposed layer at matched depths, while both contact force and commanded scrape depth approached the same approximately 0.9 mm depth limit. We also showed that penetration depth can be used to control which colour layer is revealed, and demonstrated that the ergodic planner can connect an input image to a planned manipulation mission.

There are several directions for future work. First, the physics-based solver should be calibrated against real material measurements using the existing calibration procedure, allowing its results to be quantitatively validated. Second, the planner can be extended to support trace primitives in addition to the current scrape and dot actions. Finally, further experiments with different tool geometries can determine whether the observed 0.9 mm depth ceiling is specific to the loop tool or

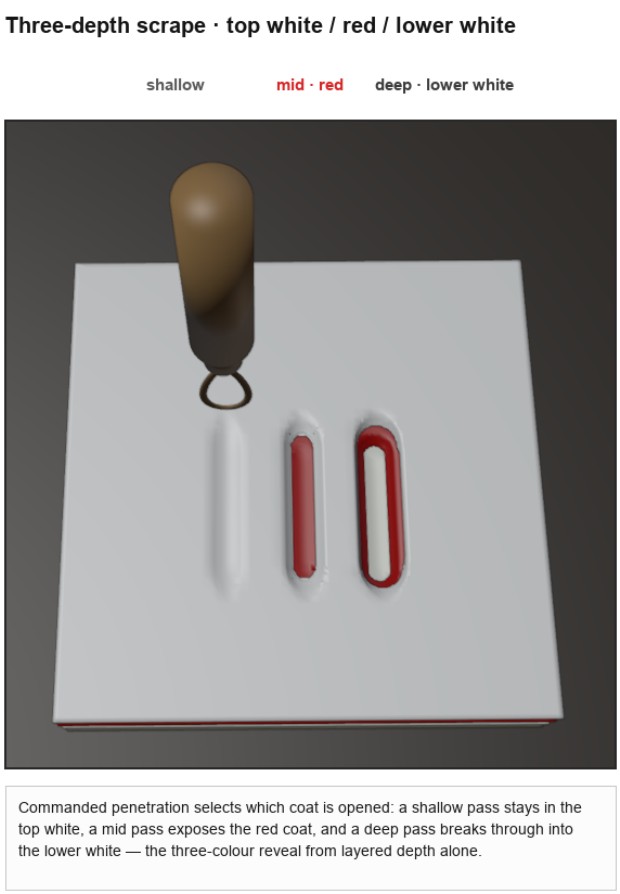


Fig. 7. Three-depth scrape exposing the top white, red, and lower white layers by commanded penetration depth alone.

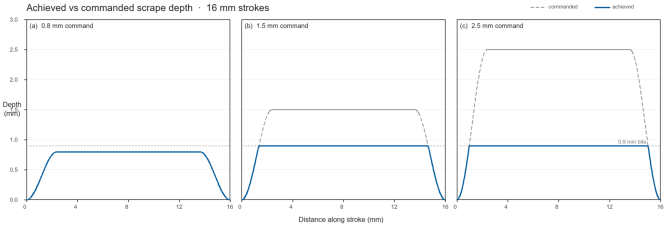


Fig. 8. Achieved versus commanded scrape depth over 16 mm strokes at three commanded depths. Achieved depth plateaus at ∼0.9 mm regardless of the command above that point.

represents a more general material-compaction limit. Once the primitive behaviours are better characterized, future work can investigate learning these primitives from demonstrations. The resulting primitives could then be composed through higher-level planning, including language-based approaches, to move toward the larger goal of translating artistic intent into robotic sgraffito actions.

Acknowledgments

The simulation and ergodic planning framework used in this project, including the canvas and tool representation, the physics-based solver, and the ergodic planner, was developed by our supervisor, Prof. SiQi Zhou. This project's contribution is the experimental design, evaluation, and analysis presented in Sections V and VI, conducted using that framework.

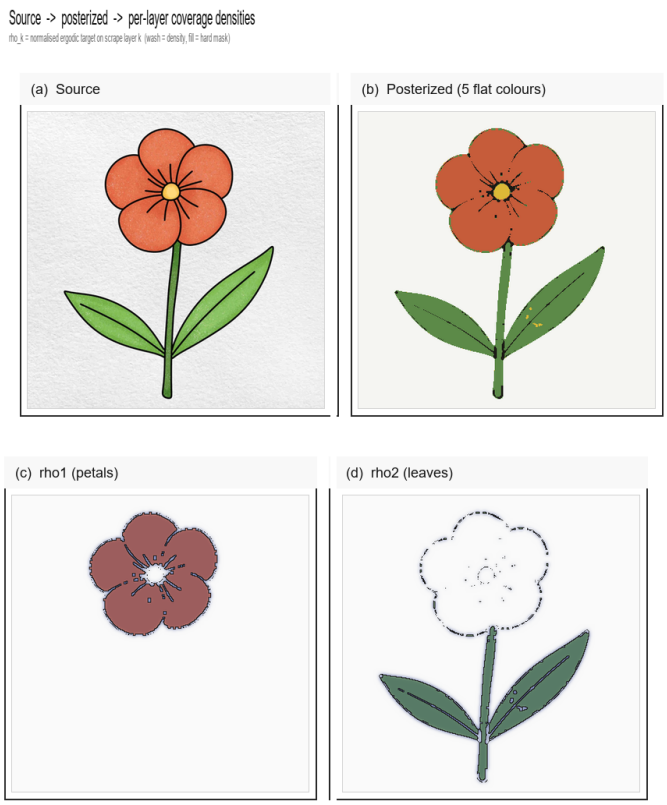


Fig. 9. Source image, posterized image, and per-layer target coverage densities (ρ_1: petals, ρ_2: leaves) used by the ergodic planner.

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